CURRICULUM VITAE

Memduh Burak Ardıç

Water Resources Engineer | Modeling Specialist, River Basin Planning & Climate Change



Location	Amstelveen (Amsterdam Area), The Netherlands
Permit	Dutch residence and work permit
Expertise	Hydrological Modelling, Water Resources Planning and Allocation, Hydrodynamical Modelilg, Water Quality Modelling, Climate Change Impact Assessment, Drought Risk Management, GIS and Remote Sensing
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Portfolio	memduhardic.github.io

SUMMARY

I am a water resources engineer who loves solving problems, especially in modelling. I have 8 years of experience in hydrological, water allocation, water quality and hydrodynamic modelling for river basin planning. Since 2023 I have been a Project Expert at NFB Engineering and Consulting on the EU-funded Technical Assistance on Preparation of River Basin Management Plans for Six Basins (6RBMP). This is the biggest river basin management project in Türkiye and the first in the country to prepare River Basin Management Plans together with Sectoral Water Allocation Plans. On this project I work within the international technical assistance team on Water Framework Directive implementation. I also work on the Kızılırmak Basin Master Plan for the State Hydraulic Works (DSİ) and on international water resources planning projects.

I have delivered modelling studies from the Terms of Reference to the final report. This includes setting up the data and model chain, calibrating models, running climate and management scenarios, writing technical reports and presenting the results to ministries, regulators and stakeholders. I also write the technical parts of proposals, including World Bank-format methodologies, work plans and person-month estimates. Alongside project work, I develop open-source Python tools that make modelling workflows faster and reproducible.

My main interests are environmental modelling, numerical modelling and climate scenario analysis. In my recent work I use integrated modelling systems to assess how climate change affects hydrological systems, water bodies and environmental dynamics. For this I work with observational datasets, spatial data and global climate model outputs to prepare model simulations. I enjoy contributing to projects that support sustainable water management, environmental risk assessment and climate adaptation strategies. I also have a growing enthusiasm for remote sensing and Earth observation, especially for hydrology, environmental problems and decision-making in water resources management.

PROFESSIONAL EXPERINCES

2026 - present	Project Consultant (remote from the Netherlands), NFB Engineering and Consulting Inc.
2023 - 2026	Project Expert, NFB Engineering and Consulting Inc., Ankara, Türkiye
2021 - 2023	Project Engineer, NFB Engineering and Consulting Inc., Ankara, Türkiye
2018 - 2021	Researcher - Graduate Student, Gazi University, Ankara, Türkiye
2018 - 2019	Project Engineer, io Environmental Solutions Research and Development Co., Ankara, Türkiye

EDUCATION

2018 - 2021	MSc Civil Engineering (Hydraulics and Water Resources Management), Gazi University, Ankara. Thesis: *Hydrodynamic and water quality modelling in tidal estuaries: the Ribble Estuary case*. GPA 3.61/4.00
2013 - 2018	BSc Civil Engineering, Gazi University, Ankara. Thesis: *Numerical analysis of seepage and flow nets by the finite difference method*

PROJECT EXPERIENCE

PERIOD
2026 - present
ROLE
Hydrological modelling expert; remote consultant from Oct 2026
COUNTRY
Türkiye (Kızılırmak Basin)
CLIENT
State Hydraulic Works (DSİ)
COMPANY
NFB Engineering and Consulting

Kızılırmak Basin Master Plan

The project Long-term water resources development plan for the basin of Türkiye's longest river, combining hydrology, climate change, remote sensing of irrigated lands, water quality and GIS data, with calibrated hydrological models to quantify water availability under current and future conditions.

Contribution:

- Hydrological analyses and rainfall-runoff modelling of the basin, from model setup to calibration and validation
- Tested seven models (SWAT+, gridded HEC-HMS, GR2M, GR4J, CemaNeige-GR4J, HBV, ANN/LSTM) on two pilot catchments to support the client's model selection, as required by the ToR
- Automatic calibration with SCE-UA and comparison by NSE, KGE, RMSE, MAE, PBIAS, R^2, water balance, run time and calibration effort
- Continuous gridded HEC-HMS model (SMA, ModClark, linear-reservoir baseflow, temperature-index snowmelt) forced by 34 long-record stations
- Defined the planning sub-basins with delineation alternatives study
- Prepared a data inventory and inception report sections on sub-basins, data requirements, previous studies and model evaluation

Tools *SWAT+, HEC-HMS, GR4J, HBV, LSTM, Python (SPOTPY), WhiteboxTools, QGIS*

PERIOD
2023 - present
ROLE
Project Expert; remote consultant from Oct 2026
COUNTRY
Türkiye (Antalya, Batı Karadeniz, Doğu Akdeniz, Doğu Karadeniz, Kızılırmak, Marmara)
BENEFICIARY
Directorate General of Water Management, Ministry of Agriculture and Forestry
FUNDING
European Union and Republic of Türkiye (IPA II)
COMPANY
NFB Engineering and Consulting

Technical Assistance on Preparation of River Basin Management Plans for Six Basins (6RBMP)

6rbmp.tarimorman.gov.tr/en

The project The biggest river basin management project in Türkiye and one of the country's largest technical assistance projects (EUR 6.9 million). It prepares River Basin Management Plans in line with the EU Water Framework Directive for six basins and is the first in Türkiye to combine them with Sectoral Water Allocation Plans, bringing the number of Turkish basins with a management plan to 18 of 25.

Contribution:

- Set up, configured and calibrated the basin-wide modelling system for hydrology, water allocation and water quality, and defined the planning scenarios
- Built the WEAP sectoral water allocation models for the Kızılırmak and Doğu Akdeniz basins (Activity 4.1.6) and presented the results to the beneficiary and stakeholders
- Integrated climate change into the modelling chain: GCM evaluation, statistical and dynamical downscaling, bias-corrected forcings, trend and scenario analysis
- Led drought characterisation and risk assessment, including the link between low flows and water quality deterioration
- Reviewed GESCAL water quality calibration for Kızılırmak and Marmara water bodies, including the effect of below-LOQ monitoring data on PBIAS
- Preliminary design of a real-time decision support system: monitoring network review, new streamflow, meteorological, reservoir and groundwater stations with cost estimates, and a SCADA-based link to the models
- Kept the modules consistent and automated large AquaTool/Excel workflows in Python; used Sentinel-1/2, ERA5 and MODIS data for model inputs and machine learning
- Presented model results to the client, regulators and stakeholders; joined field investigations to validate model assumptions

Tools *WEAP, AquaTool (EVALHID, SIMGES, GESCAL), Python, Sentinel-1/2, ERA5, MODIS*

PERIOD 2024 - 2025 ROLE Hydrological modelling expert COUNTRY Georgia CLIENT Ministry of Environmental Protection and Agriculture of Georgia (MEPA) COMPANY NFB Engineering and Consulting	**Dam Safety Assessment Study for Five Dams in Georgia** **The project** Safety assessment of the Khorkhori, Mtis Dziri, Yakublo, Pantiani and Narekvavi dams, contracted together with a socio-economic feasibility study for the Ilto Reservoir, to secure irrigation water supply to agricultural land. **Contribution:** • Hydrological modelling of the dam catchments for the safety assessment • Climate change analysis with Regional Climate Model (RCM) outputs to evaluate future hydrological conditions at the dams
PERIOD 2021 - 2023 ROLE Project Engineer COUNTRY Türkiye (Yeşilırmak, Batı Karadeniz, Sakarya, Susurluk) CLIENT Directorate General of Water Management (DGWM) COMPANY NFB Engineering and Consulting	**Drought Management Plans for the Yeşilırmak, Batı Karadeniz, Sakarya and Susurluk Basins** **The project** Basin-scale drought management plans assessing meteorological, hydrological and agricultural drought, water availability and sectoral water use, vulnerabilities, and mitigation measures and action plans. **Contribution:** • Coordinated the technical reports: water budgets, drought analysis, sectoral vulnerability analysis and measure plans • Set up GR2M and HEC-HMS models to estimate current and future water potential under GCM projections • Computed basin-scale drought indices (SPI, SPEI, Palmer indices) for historical and future periods • Led the sectoral vulnerability analysis (agriculture, industry, drinking water, ecosystems, tourism, energy, health) and the drought mitigation measures • Presented to the client and stakeholders; trained colleagues in hydrological and hydrodynamic modelling; guest lecture on HEC-HMS at Ankara Yıldırım Beyazıt University **Tools** *GR2M, HEC-HMS, ArcGIS, QGIS, Python, R, C++*
PERIOD 2018 - 2021 ROLE Researcher (MSc) COUNTRY United Kingdom (Ribble Estuary) INSTITUTION Gazi University	**Hydrodynamic and Water Quality Modelling of the Ribble Estuary** **The project** MSc research on tidal hydrodynamics, sediment transport and bacterial water quality in a macro-tidal estuary. **Contribution:** • 3-D hydrodynamic modelling of tidal forcing and wetting-drying areas; testing of numerical schemes, grid refinement and turbulence models • Wrote Navier-Stokes and advection-diffusion solvers in Python and FORTRAN • Water quality modelling of faecal coliform distribution driven by the hydrodynamic outputs **Tools** *Delft3D, D-Water Quality, DIVAST, Python, FORTRAN*
PERIOD 2018 - 2019 ROLE Project Engineer COUNTRY Türkiye (Seyhan, Ceyhan, Asi) COMPANY io Environmental Solutions R&D Co.	**Drought Management Plans for the Seyhan, Ceyhan and Asi Basins** **The project** Basin-scale drought management planning for three basins in southern Türkiye. **Contribution:** • Hydrological modelling for the three basins, including preparation of hydrological and meteorological inputs • Processed field data for the water budget studies and contributed to technical reporting

PERIOD Ongoing ROLE Technical proposal writer COUNTRY International COMPANY NFB Engineering and Consulting	**Proposal Development for National and International Tenders** **The project** Technical parts of NFB's proposals for consultancy assignments in Türkiye and abroad, including World Bank-funded projects. **Contribution:** • Wrote World Bank-format technical forms: comments on the ToR, approach, methodology and work plan, and work schedule and deliverables • Translated ToR scope into tasks, deliverables and a work programme, and estimated key and non-key expert inputs in person-months

CODING PROJECTS

TYPE Open-source tool (MIT) BUILT WITH Python, PySide6	**StationGrid - gridded meteorological forcing for HEC-HMS** github.com/memduhardic/stationgrid **What it does** Interpolates station precipitation and temperature onto watershed grids and exports CF-1.8 compliant NetCDF for gridded hydrological models; built for the Kızılırmak modelling work. **Highlights** • Vectorised IDW, automatic UTM detection and basin-masked grids • ModClark grid-cell and basin-average exports; interactive map viewer
TYPE Open-source tool (MIT) BUILT WITH Python, scikit-learn, PyTorch	**Rainfall-Runoff Benchmark - conceptual vs. machine-learning models** github.com/memduhardic/rainfall-runoff-benchmark **What it does** Compares GR4J and HBV (with snow) against Random Forest, ANN and LSTM on the same catchment, with identical inputs and the same independent test period. **Highlights** • Chronological train/validation/test split with an unseen test period; SCE-UA calibration • Configurable models, objective function and basin data; outputs load directly into Calibration Studio
TYPE Open-source tool (MIT) BUILT WITH Python, PySide6, Matplotlib	**Calibration Studio - multi-model calibration and evaluation** github.com/memduhardic/calibration-studio **What it does** Evaluates any number of model outputs (e.g. SWAT, HEC-HMS, GR4J, HBV, LSTM) against observed streamflow in one place. **Highlights** • NSE, KGE, RMSE, MAE, PBIAS, r, R^2 and volume ratio per calibration and validation period • Report-ready hydrographs with precipitation; CSV/PNG export and saved figure styling
TYPE Research study AREA Batı Karadeniz Basin, Türkiye	**Machine-learning prediction of drought indices under climate change** **What it does** Random Forest and Gradient Boosting models predicting 29 drought indices (SPI, SPEI, RDI, PNPI, Palmer family, deciles) for five sub-basins from precipitation, PET, temperature and streamflow. **Highlights** • Projections to 2099 with RegCM4.3-downscaled HadGEM2-ES, MPI-ESM-MR and CNRM-CM5 (RCP4.5/8.5) • Data leakage checks; evaluation with R^2, MAE, RMSE and Taylor diagrams

TECHNICAL SKILLS

Hydrological Modeling	HEC-HMS, SWAT, HBV, GR2M, GR4J, Temez, ANN/LSTM
Basin Modeling	WEAP, AquaTool (GESCAL)
Hydrodynamical Modeling	Delft3D, SWAN, DIVAST, HEC-RAS, SOBEK
Water Quality Modeling	D-WAQ, AquaTool (GESCAL)

Programming	Python (pandas, NumPy, xarray, scikit-learn, PyTorch, PySide6), R, MATLAB, FORTRAN, C++, Git
GIS & remote sensing	ArcGIS, QGIS, WhiteboxTools, GeoPandas; Sentinel-1/2, ERA5, MODIS; NetCDF/CF

LANGUAGES

Turkish Native

English Professional Efficiency

IELTS Academic 6.5 (British Council, 2025).

WORKSHOPS AND TRAININGS

Date	Event	Place	My role
Sep 2026	Sectoral Water Allocation Plans, Kızılırmak and Doğu Akdeniz Basins (6RBMP)	Türkiye	Presented WEAP-based allocation results
Jan 2025	Training Workshops on Modelling, Environmental Objectives and Programme of Measures II	Ankara	Speaker: climate change approaches, GCMs and RCMs
Nov 2024	Improving Public and Sector Representatives' Awareness on Water Allocation	Ankara	Speaker: allocation strategies during drought
Sep 2024	Training Workshops on Modelling, Environmental Objectives and Programme of Measures I	Ankara	Speaker: climate change in modelling
2021	Guest lecture, Ankara Yıldırım Beyazıt University	Ankara	Lecturer: HEC-HMS applications
2021 - 2023	Internal training on hydrological and hydrodynamic modelling (NFB)	Ankara	Trainer

PUBLICATIONS

Accepted

Kacikoc, M., Mesta, B., **Ardıç, M.B.**, Yalcin, A., Turan, B. Developing a comparative decision support framework through AQUATOOL and WEAP integration: Case of Manavgat Basin. Journal of Water Resources Planning and Management.

Conference papers

Ardıç, M.B., Turan, B., Turan, B., 2026. Determination of flow duration curve-based drought scenarios in modeling studies. 13th National Hydrology Congress, Gebze Technical University, Kocaeli, Türkiye (full paper).

Koçyiğit, Ö., **Ardıç, M.B.**, 2021. Investigation of a type of the local grid refinement in the model harbour. Middle East International Conference on Contemporary Scientific Studies-V, Ankara, Vol. 1, No. 1, pp. 548-557.

Theses

Ardıç, M.B., 2021. Hydrodynamic and water quality modelling in tidal estuaries: the Ribble Estuary case. MSc thesis, Gazi University.

Ardıç, M.B., 2018. Numerical analysis of seepage and flow nets by the finite difference method. BSc thesis, Gazi University.

CERTIFICATIONS

2019	Methodologies and Applications in Project Management, Project Management Institute (PMI), Ankara

REFERENCES

Dr. Önder Koçyiğit, MSc and BSc thesis supervisor, Gazi University - konder@gazi.edu.tr, +90 312 582 32 12

Assoc. Prof. Müsteyde Baduna Koçyiğit, Gazi University - baduna@gazi.edu.tr, +90 312 582 32 23